



Dolby[®] Media Decoder User's Manual

15 October 2015

Software v2.0

Dolby Laboratories, Inc.

Corporate Headquarters

Dolby Laboratories, Inc.

100 Potrero Avenue

San Francisco, CA 94103-4813 USA

Telephone 415-558-0200

Fax 415-863-1373

www.dolby.com

European Headquarters

Dolby Laboratories, Inc.

Wootton Bassett

Wiltshire SN4 8QJ England

Telephone 44-1793-842100

Fax 44-1793-842101

DISCLAIMER OF WARRANTIES:

PRODUCTS MANUFACTURED BY DOLBY ARE WARRANTED AGAINST DEFECTS IN MATERIALS AND WORKMANSHIP FOR A PERIOD OF ONE YEAR FROM THE DATE OF PURCHASE. IN THE CASE OF SOFTWARE, THE MEDIA ON WHICH THE SOFTWARE IS FURNISHED IS WARRANTED AGAINST DEFECTS IN MATERIALS AND WORKMANSHIP FOR A PERIOD OF 60 DAYS FROM THE DATE OF PURCHASE. ALL OTHER WARRANTIES, CONDITIONS AND OTHER TERMS (WHETHER EXPRESS OR IMPLIED), INCLUDING, BUT NOT LIMITED TO, THOSE RELATING TO NON-INFRINGEMENT OF THIRD-PARTY RIGHTS (INCLUDING, BUT NOT LIMITED TO, PATENT AND COPYRIGHT RIGHTS), SATISFACTORY QUALITY, MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE ARE, TO THE FULLEST EXTENT PERMITTED BY LAW, EXCLUDED FROM THESE EQUIPMENT TERMS OF SALE.

LIMITATION OF LIABILITY:

IT IS UNDERSTOOD AND AGREED THAT DOLBY LABORATORIES' LIABILITY, WHETHER IN CONTRACT, IN TORT, UNDER ANY WARRANTY, IN NEGLIGENCE, OR OTHERWISE, SHALL NOT EXCEED THE COST OF REPAIR OR REPLACEMENT OF THE DEFECTIVE COMPONENTS OR ACCUSED INFRINGING DEVICES, AND UNDER NO CIRCUMSTANCES SHALL DOLBY LABORATORIES BE LIABLE FOR INCIDENTAL, SPECIAL, DIRECT, INDIRECT, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, DAMAGE TO SOFTWARE OR RECORDED AUDIO OR VISUAL MATERIAL), COST OF DEFENSE, OR LOSS OF USE, REVENUE, OR PROFIT, EVEN IF DOLBY LABORATORIES OR ITS AGENTS HAVE BEEN ADVISED, ORALLY OR IN WRITING, OF THE POSSIBILITY OF SUCH DAMAGES.

Dolby, Dolby Atmos, Pro Logic, and the double-D symbol are registered trademarks of Dolby Laboratories.

Dolby Digital Plus, MLP Lossless, and Surround EX are trademarks of Dolby Laboratories.

All other trademarks remain the property of their respective owners.

© 2015 Dolby Laboratories. All rights reserved.

Table of Contents

List of Figures	v
List of Tables.....	vii
Chapter 1 Introduction to the Dolby Media Decoder User's Manual	
1.1 About This Documentation	1
1.2 Abbreviated Speaker Notations.....	1
1.3 Contacting Dolby	3
Chapter 2 Installing and Launching Dolby Media Decoder	
2.1 Launching Dolby Media Decoder	5
2.1.1 Quitting Dolby Media Decoder	5
2.2 Managing Dolby Media Decoder Windows.....	6
2.2.1 Latching and Unlatching Dolby Media Decoder Windows	6
2.2.2 Closing and Opening Dolby Media Decoder Windows	7
2.3 Checking the Software Version Number.....	7
Chapter 3 Setting up Dolby Media Decoder	
3.1 Setting Up Internal or External Audio Hardware.....	9
3.2 Setting Speaker Configurations.....	10
3.2.1 Creating Listening Modes	11
3.2.2 Selecting Output Channel Locations.....	12
3.2.3 Device	15
3.2.4 Output	15
3.2.5 Size.....	16
3.2.6 Trim.....	16
3.2.7 Delay.....	16
3.2.8 Subwoofer Crossover	16
3.2.9 Dolby Pro Logic Delay	16
3.2.10 Testing a Listening Mode	16
3.3 Changing Monitor Control Settings.....	17
3.3.1 Dolby Pro Logic II.....	18
3.3.2 Decode Controls	18
3.3.3 Custom Compression	18
3.4 Dolby Headphone Output	18
Chapter 4 Using Dolby Media Decoder	
4.1 Using the Transport Window	19
4.1.1 Loading Audio Files to Play Back	20
4.1.2 Setting the Display Type	21
4.1.3 Setting the Display Mode	22
4.2 Setting Up Video for Synchronous Playback via a QuickTime Movie.....	22
4.3 Using the Transport Controls.....	24
4.3.1 Determining the Device Controlled	24

4.4	Monitoring Playback	25
4.4.1	Meters/Gain Window.....	25
4.4.2	Monitor Window	26
4.4.3	Buffer Underflow Indicator	29
 Chapter 5 Working with Decoder Configuration Files		
5.1	Saving, Loading, and Deleting Decoder Configuration Files	31
 Chapter 6 Viewing Metadata		
6.1	The Metadata Window.....	33
6.2	Viewing Metadata in the Metadata Window	34
 Chapter 7 Application Menus		
7.1	Dolby Media Decoder Menu	35
7.2	File Menu	36
7.3	Edit Menu	36
7.4	Transport Menu	37
7.5	Monitoring Menu	37
7.6	Window Menu	38
7.7	Help Menu	39
 Chapter 8 Dolby Media Decoder Frequently Asked Questions		
8.1	How Can I Get My Pro Tools I/O to Appear in the Output Device Selection Drop-Down Menu on the Speaker Settings Page?.....	41
8.2	How Do I Get Audio from All of the Channels of My Pro Tools I/O?.....	41
8.3	Why Does Dolby Media Decoder Not Recognize My I/O Device?	41
8.4	Why Am I Not Hearing Audio?.....	42
8.5	Why Does the Audio Sound Incorrect?	42
 Chapter 9 Reference Information		
9.1	Dolby Professional Technologies	43
9.2	Dolby Atmos	43
9.3	Dolby TrueHD.....	44
9.3.1	Features.....	44
9.3.2	Benefits.....	44
9.3.3	Compatibility with Current A/V Receivers	44
9.4	Dolby Digital Plus	44
9.4.1	Features.....	45
9.4.2	Benefits.....	45
9.5	Dolby Digital	45
9.6	MLP Lossless	46
 Chapter 10 Calibrating the Speaker System		
10.1	Speaker Calibration Functionality	47
10.2	Calibrating Speaker Playback Levels	48
10.2.1	Measuring SPL	49
10.2.2	Subwoofer Calibration	50
10.3	Calculating Speaker Delay Values	51

List of Figures

Figure 2-1	Dolby Media Decoder Icon	5
Figure 2-2	Transport, Meters/Gain, and Monitor Windows	5
Figure 3-1	Default Speaker Configuration in the Speaker Settings Page	10
Figure 3-2	Example Settings for a Listening Mode	11
Figure 3-3	The Monitor Window.....	12
Figure 3-4	Relative Positions of Floor Speakers (Not to Scale).....	14
Figure 3-5	Relative Positions of Overhead Speakers (Not to Scale)	15
Figure 3-6	Monitor Control Page in the Preferences Window.....	17
Figure 4-1	Transport Window	19
Figure 4-2	Information Display in the Transport Window.....	21
Figure 4-3	File Position Displayed in Timecode (23.976 fps)	21
Figure 4-4	QuickTime Movie Screen	23
Figure 4-5	Remote and Sync Buttons.....	24
Figure 4-6	Meters/Gain Window with Updated Metering	25
Figure 4-7	The Monitor Window.....	26
Figure 4-8	Bypass Button	27
Figure 4-9	Buffer Underflow Indicator in the Transport Window	29
Figure 5-1	Example Configurations Page in the Preferences Window	31
Figure 5-2	Example Misconfiguration Warning	32
Figure 5-3	Missing Device Indications on the Speaker Settings Page	32
Figure 6-1	Metadata Window for Dolby Digital Plus Bitstream with Dolby Atmos	33
Figure 7-1	Dolby Media Decoder Menu	35
Figure 7-2	File Menu.....	36
Figure 7-3	Edit Menu	36
Figure 7-4	Transport Menu	37
Figure 7-5	Monitoring Menu	38
Figure 7-6	Window Menu.....	38
Figure 7-7	Help Menu	39
Figure 10-1	Real-Time Analyzer Display Simulation	50

List of Tables

Table 3-1	Channel Location Labels and Descriptions	12
Table 4-1	Decoder Transport Controls	24
Table 8-1	Single-Detail Problems and Diagnoses	42

Introduction to the Dolby Media Decoder User's Manual

This manual describes how to use the Dolby® Media Decoder.

1.1 About This Documentation

This manual is for content professionals who need to decode and monitor Dolby audio bitstreams that were created with Dolby Media Encoder from standard channel-based audio or a Dolby Atmos® master file.

Dolby Media Decoder supports the following encoded formats:

- Dolby TrueHD bitstream with Dolby Atmos
- Dolby Digital Plus™ bitstream with Dolby Atmos
- Dolby TrueHD bitstream
- Dolby Digital Plus bitstream
- Dolby Digital bitstream
- MLP Lossless™ bitstream

The Dolby Media Decoder supports monitoring and auditioning of consumer listening modes, including downmixing and dynamic range control.

1.2 Abbreviated Speaker Notations

The *Dolby Media Decoder User's Manual* documentation uses specific abbreviations for each speaker channel.

This table lists Dolby abbreviated speaker notations and corresponding speaker notations for other specifications. An em dash (—) indicates that no corresponding notation exists.

Table 1-1 Abbreviated Speaker Notations

Dolby Speaker Name	Dolby Abbreviation	CEA-861 Abbreviation	ITU-R BS.2159-5 (Type B 10.2 Channel) Abbreviation	SMPTE 2036-2 (22.2 Channel) Abbreviation	IEC 62574 (30.2 Channel) Abbreviation
Left	L	FL	L	FL	FL
Right	R	FR	R	FR	FR
Center	C	FC	C	FC	FC
Left Surround	Ls	RL	LS	—	LS and LSd
Right Surround	Rs	RR	RS	—	RS and RSd
Left Rear Surround/Left Back	Lrs	RLC	LB	BL	BL

Table 1-1 Abbreviated Speaker Notations

Dolby Speaker Name	Dolby Abbreviation	CEA-861 Abbreviation	ITU-R BS.2159-5 (Type B 10.2 Channel) Abbreviation	SMPTE 2036-2 (22.2 Channel) Abbreviation	IEC 62574 (30.2 Channel) Abbreviation
Right Rear Surround/Right Back	Rrs	RRC	RB	BR	BR
Center Surround	Cs	RC	—	BC	BC
Left Wide	Lw	FLW	—	—	FLW
Right Wide	Rw	FRW	—	—	FRW
Left Center	Lc	FLC	—	FLC	FLC
Right Center	Rc	FRC	—	FRC	FRC
Left Surround 1	Ls1	—	—	SiL	SiL
Right Surround 1	Rs1	—	—	SiR	SiR
Left Front Height	Lfh	FLH	LH	TpFL	TpFL
Right Front Height	Rfh	FRH	RH	TpFR	TpFR
Left Top Middle	Ltm	—	—	—	TpLS
Right Top Middle	Rtm	—	—	—	TpRS
Left Top Front	Ltf	—	—	TpSiL	TpSiL
Right Top Front	Rtf	—	—	TpSiR	TpSiR
Left Top Rear	Ltr	—	—	—	—
Right Top Rear	Rtr	—	—	—	—
Left Rear Height	Lrh	—	—	TpBL	TpBL
Right Rear Height	Rrh	—	—	TpBR	TpBR
Low-Frequency Effects 1	LFE/LFE1	LFE	LFE1	LFE1	LFE1
Left Screen	Lsc	—	—	—	—
Right Screen	Rsc	—	—	—	—
Left Surround 2	Ls2	—	—	—	—
Right Surround 2	Rs2	—	—	—	—
Left Rear Surround 1	Lrs1	—	—	—	—
Right Rear Surround 1	Rrs1	—	—	—	—
Left Rear Surround 2	Ls2	—	—	—	—
Right rear Surround 2	Rs2	—	—	—	—
Left Center Surround	Lcs	—	—	—	—
Right Center Surround	Rcs	—	—	—	—
Center Height	Ch	FCH	—	TpFC	TpFC
Top Surround	Ts	TC	—	TpC	TpC
Left Binaural	Lbin	—	—	—	—
Right Binaural	Rbin	—	—	—	—
Left Only	Lo	—	—	—	—
Right Only	Ro	—	—	—	—
Left Total	Lt	—	—	—	—
Right Total	Rt	—	—	—	—
Left Sound Direct	Lsd	—	—	—	—
Right Sound Direct	Rsd	—	—	—	—
Left Dolby Enabled	Le	—	—	—	—
Right Dolby Enabled	Re	—	—	—	—
Left Surround Dolby Enabled	Lse	—	—	—	—

Table 1-1 Abbreviated Speaker Notations

Dolby Speaker Name	Dolby Abbreviation	CEA-861 Abbreviation	ITU-R BS.2159-5 (Type B 10.2 Channel) Abbreviation	SMPTE 2036-2 (22.2 Channel) Abbreviation	IEC 62574 (30.2 Channel) Abbreviation
Right Surround Dolby Enabled	Rse	—	—	—	—
Left Rear Surround Dolby Enabled	Lrse	—	—	—	—
Right Rear Surround Dolby Enabled	Rrse	—	—	—	—

1.3 Contacting Dolby

You can contact Dolby regarding this product and its supporting documentation.

If you have technical questions about this product, please contact dmpSupport@dolby.com.

If you have questions or comments about this document, please contact [technical publications](#).

Installing and Launching Dolby Media Decoder

The Dolby® Media Decoder application is part of the Dolby Media Producer Suite. It is installed when the full suite is installed. Refer to the Dolby Media Producer Suite installation guide or what's new documentation for system requirements and installation steps.

2.1 Launching Dolby Media Decoder

The Dolby Media Decoder application provides access to all Dolby Media Decoder user-facing controls.

The Dolby Media Decoder client application icon is labeled Dolby Media Decoder.



Figure 2-1 Dolby Media Decoder Icon

To launch the application, do one of the following:

- Click the Dolby Media Decoder application icon.
- Double-click the Dolby Media Decoder application file (located in /Applications/Dolby/Dolby Media Producer/).

The three main Dolby Media Decoder windows appear (latched together) in the following order: transport, meters/gain, and monitor.

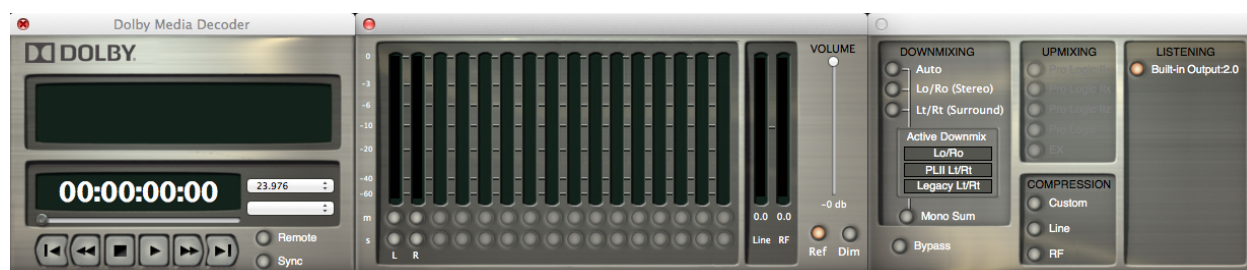


Figure 2-2 Transport, Meters/Gain, and Monitor Windows

2.1.1 Quitting Dolby Media Decoder

You can quit the Dolby Media Decoder session from the Dolby Media Decoder menu or icon.

To quit the application, do one of the following:

- In the Dolby Media Decoder menu bar, choose **Dolby Media Decoder > Quit Dolby Media Decoder**.
- Press <Command>+<Q>.
- Right-click the Dolby Media Decoder icon, and choose **Quit**.

2.2 Managing Dolby Media Decoder Windows

2.2.1 Latching and Unlatching Dolby Media Decoder Windows

Dolby Media Decoder launches by default with the three main windows (latched together), in the following order: transport, meters/gain, and monitor. You can arrange the windows in a different sequence by unlatching and moving them (or latching them in a different sequence).

Unlatching One Window

You can unlatch one window so that it can be moved around the screen independently.

To unlatch a window:

1. Click the window you want to unlatch.
2. Choose **Window > Unlatch**.

Unlatching All Latched Windows

You can unlatch all windows so that they can be moved around the screen independently.

To unlatch all latched windows:

- Choose **Window > Unlatch All**.

Latching Unlatched Windows

When one or two windows are not latched, you can latch two of the open windows. The unlatched window will latch to the right-side edge of the other two windows if those two are already latched, or it will latch to the right edge of the transport window if all three are floating around.

To latch an unlatched window:

- Click the window you want to latch.
- Choose **Window > Latch**.

Latching All Three Open Windows

You can latch all three windows in their default order with the transport window on the left, the meters/gain window in the center, and the monitor window on the right.

To latch all three windows along a vertical edge:

- Choose **Window > Latch All**.

2.2.2 Closing and Opening Dolby Media Decoder Windows

Dolby Media Encoder provides windows buttons and a menu command to close or open its windows.

To close a Dolby Media Decoder window:

- Click the button in the upper-left corner of the window.
- From the menu bar, choose **Window**, and then one of the window names in the menu. When latched, the transport, monitor, and meters/gain windows cannot be closed by using the **Window** menu command.

To open a Dolby Media Decoder window:

- From the menu bar, choose **Window**, and then one of the window names in the menu.

2.3 Checking the Software Version Number

You can check the version number of the Dolby Media Decoder, including the version number of the parent Dolby Media Producer Suite.

To check the version number of the Dolby Media Decoder software application:

- Choose **Dolby Media Decoder > About Dolby Media Decoder**.
- Click the Dolby logo in the top-left corner of the transport window.

Setting up Dolby Media Decoder

When Dolby® Media Decoder is installed, it is initially configured to use the built-in line-level outputs to decode two-channel audio to the speakers attached to the local machine. If you wish to use external hardware or more than two speakers, you must set up a new speaker configuration for the decoder.

To set up the application:

1. Set up your audio output, either internal or external (see [Section 3.1](#)).
2. Set up your speaker configurations (see [Section 3.2](#)).
3. Test the results (see [Section 3.2.10](#)).



Tip: In addition to a proper application setup, a carefully calibrated listening environment is crucial to proper content evaluation. You should need to do this only once for each desired mode; however, we recommend that you perform a quick check before each session to reconfirm the correct settings.

See [Chapter 10](#) for instructions for calibrating speaker playback levels, channel delays, and bass management.

4. Optional: Configure custom compression options. (see [Section 3.3.3](#)).
5. When you are satisfied with your monitoring configurations, you may want to save them as a decoder configuration file. See [Chapter 5](#).

3.1 Setting Up Internal or External Audio Hardware

Dolby Media Decoder automatically detects built-in audio hardware and most audio devices attached to the computer. These include Thunderbolt, FireWire®, and USB interfaces. Occasionally, some setup work is necessary before the Dolby Media Decoder recognizes an I/O device.

Consider the following when working with external audio hardware:

- Make sure that the I/O driver (if required) is properly installed. For the latest drivers, refer to the manufacturer's website.
- If the driver is properly installed, but is not recognized, reboot your computer.
- Refer to your I/O documentation, as needed. Always make sure you are using the latest firmware and driver (if needed) for your audio hardware.
- Many audio hardware devices provide a number of settings and controls, including hardware setup options and output volume control, which can affect audio playback from Dolby Media Decoder. Check these settings carefully.
- Make sure the audio device is powered on before launching Dolby Media Decoder.
- If using an Avid® I/O as your audio hardware, make sure that Pro Tools® is not currently using the Avid I/O before attempting to access it from the Dolby Media Decoder player.

3.2 Setting Speaker Configurations

Use the **Speaker Settings** page of the **Preferences** window to configure and manage as many as eight distinct user-named listening modes. Each mode can accommodate 1 to 16 channels.

By default, when you first launch the decoder, the speaker configuration will be set to stereo using the built-in output of your computer, so that the decoder is ready for immediate use.



Figure 3-1 Default Speaker Configuration in the Speaker Settings Page

All user modes can be assigned to the same I/O device to feed the same speakers, or any number of modes could be assigned to different I/O devices to feed different sets of speakers. For example, you might set up three modes: a Dolby Atmos® mode that feeds a set of 16 speakers to audition bitstreams carrying Dolby Atmos audio and metadata, a 5.1-channel mode that feeds a set of just six large speakers using the same external audio I/O, and a two-channel mode using a separate external I/O that feeds a pair of small stereo speakers.

Each mode has separate level, delay, and bass management settings. Each mode will also have a custom name assigned by you. Once modes have been saved on the **Speaker Settings** page in the **Preferences** window, they will then appear in the **LISTENING** section of the monitor window.

To open the **Speaker Settings** page:

1. Choose **Dolby Media Decoder > Preferences**, or press <Command>+<comma key>.
2. In the **Preferences** window, click the **Speaker Settings** tab.

The **Speaker Settings** page displays.

3.2.1 Creating Listening Modes

You can create a custom listening mode that will be available in the **LISTENING** section of the monitor window.

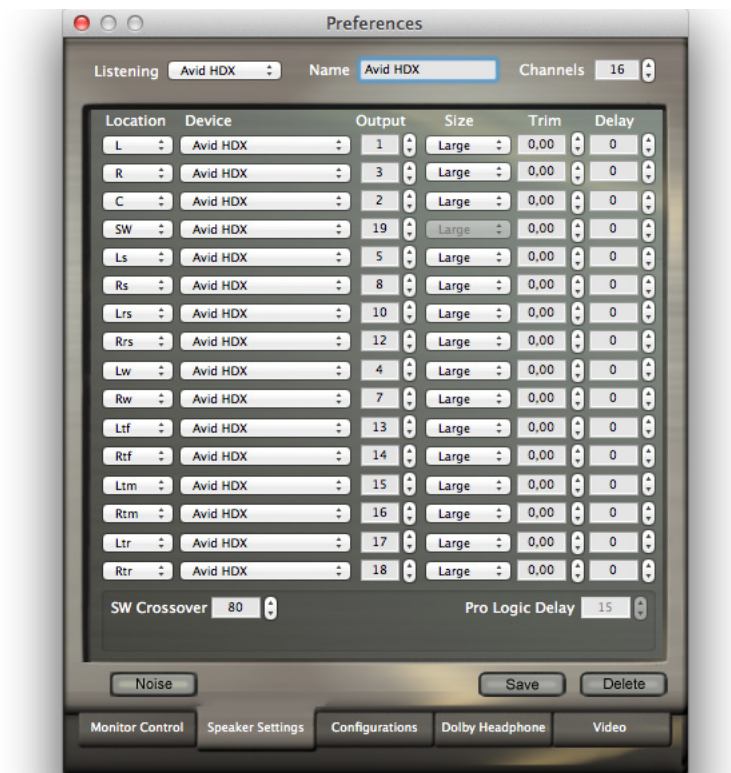


Figure 3-2 Example Settings for a Listening Mode

To create a listening mode:

1. Open the **Speaker Settings** page.
2. In the **Listening** drop-down menu, choose an unused slot.
3. In the **Channels** field, select the number of output channels required (2–16).
L and R are required. A subwoofer (SW) output must be added if defining additional channels besides L and R.
4. Select a **Location**, **Device**, **Output**, **Size**, **Trim**, and **Delay** value for each listed channel.
5. For modes using a subwoofer, select a **SW Crossover** value.
6. In the **Name** field, enter a name for the mode.
This name will appear as the button name in the **LISTENING** section of the monitor window after you save the listening mode.
7. Click the **Save** button.
This saves the configuration and places a new Listening mode button in the **LISTENING** section of the monitor window.



Figure 3-3 The Monitor Window

3.2.2 Selecting Output Channel Locations

Selecting an output channel from the **Location** drop-down menu assigns a specific rendered audio signal to a specific I/O output, such as Left (L) to output channel 1, and Right (R) to output channel 2.

Refer to this table for standard labels for the supported speaker channel locations and their descriptions.

Table 3-1 Channel Location Labels and Descriptions

Location Label	Description
L	Left, a loudspeaker position behind the screen to the far-left edge.
R	Right, a loudspeaker position behind the screen to the far-right edge.
C	Center, a loudspeaker position behind the screen, corresponding to the horizontal center of the screen.
SW	Subwoofer, a band-limited low-frequency-only loudspeaker positioned at the screen end of the room. Delivers LFE content, plus any redirected bass from bass-managed (small) speakers.
Ls	Left Surround, a loudspeaker (or an array in a 5.1 or 7.1 cinema system) along the left side of the room.
Rs	Right Surround, a loudspeaker (or an array in a 5.1 or 7.1 cinema system) along the right side of the room.
Lrs	Left Rear Surround, a loudspeaker (or an array in a 5.1 or 7.1 cinema system) on the back wall of the room in the left corner.
Rrs	Right Rear Surround, a loudspeaker (or an array in a 5.1 or 7.1 cinema system) on the back wall of the room in the right corner.
Cs	Center Surround, a loudspeaker centered on the back wall of the room.
Lc	Left Center, a loudspeaker position midway between the center of the screen and the left edge of the screen.
Rc	Right Center, a loudspeaker position midway between the center of the screen and the right edge of the screen.
Lw	Left Wide, a loudspeaker position outside the screen area, at the far-left front in the room.
Rw	Right Wide, a loudspeaker position outside the screen area, at the far-right front in the room.
Lfh	Left Front Height, an overhead loudspeaker for Dolby Atmos, at the left front of the room; Lvh is treated as Lfh.
Rth	Right Top Height, an overhead loudspeaker for Dolby Atmos, at the right front of the room; Rvh is treated as Rth.

Table 3-1 Channel Location Labels and Descriptions (continued)

Location Label	Description
Ltf	Left Top Front, an overhead loudspeaker for Dolby Atmos, at the left front of the room.
Rtf	Right Top Front, an overhead loudspeaker for Dolby Atmos, at the right front of the room.
Ltm	Left Top Mid, an overhead loudspeaker for Dolby Atmos, at the left middle of the room.
Rtm	Right Top Mid, an overhead loudspeaker for Dolby Atmos, at the right middle of the room.
Ltr	Left Top Rear, an overhead loudspeaker for Dolby Atmos, at the left rear of the room.
Rtr	Right Top Rear, an overhead loudspeaker for Dolby Atmos, at the right rear of the room.
Lrh	Left Rear Height; Dolby Digital Plus bitstreams carrying Dolby Atmos audio and associated metadata also support Lfh/Rfh (or Lvh/Rvh) left/right front height speaker locations. These locations are also supported by Dolby Digital, Dolby Digital Plus, and Dolby True HD bitstreams.
Rrh	Right Rear Height; Dolby Digital Plus bitstreams carrying Dolby Atmos audio and associated metadata also support Lfh/Rfh (or Lvh/Rvh) left/right front height speaker locations. These locations are also supported by Dolby Digital, Dolby Digital Plus, and Dolby True HD bitstreams.
Lsc	Left Surround Center, a loudspeaker along the middle of left side of the room.
Rsc	Right Surround Center, a loudspeaker along the middle of the right side of the room.
Ls1	Left Surround 1, a loudspeaker along the left side of the room.
Rs1	Right Surround 1, a loudspeaker along the right side of the room.
Ls2	Left Surround 2, a loudspeaker along the left side of the room.
Rs2	Right Surround 2, a loudspeaker along the right side of the room.
Lrs1	Left Rear Surround 1, a loudspeaker on the back wall of the room in the left corner.
Rrs1	Right Rear Surround 2, a loudspeaker on the back wall of the room in the right corner.
Lrs2	Left Rear Surround 2, a loudspeaker on the back wall of the room in the left corner.
Rrs2	Right Rear Surround 2, a loudspeaker on the back wall of the room in the right corner.
Lcs	Left Center Surround, a loudspeaker along the center of the left side of the room.
Rcs	Right Center Surround, a loudspeaker along the center of the right side of the room.

The following illustrations provide relative positioning of floor and overhead speakers. Distances and angle are not to scale. To find detailed speaker layout guidance for both traditional and Dolby Atmos home theaters, refer to the information provided on the Dolby website: <http://www.dolby.com/us/en/guide/speaker-setup/index.html>.

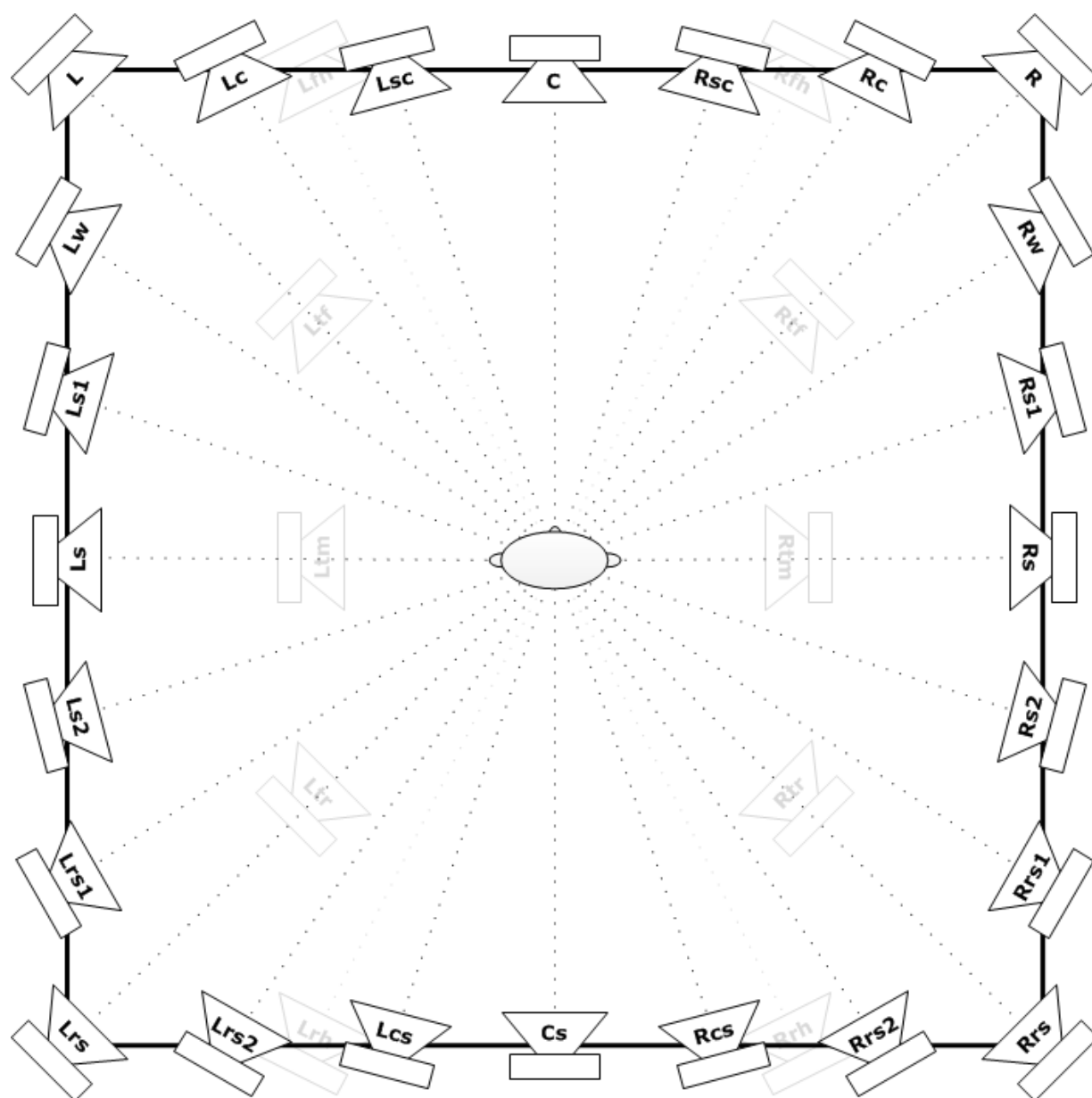


Figure 3-4 Relative Positions of Floor Speakers (Not to Scale)

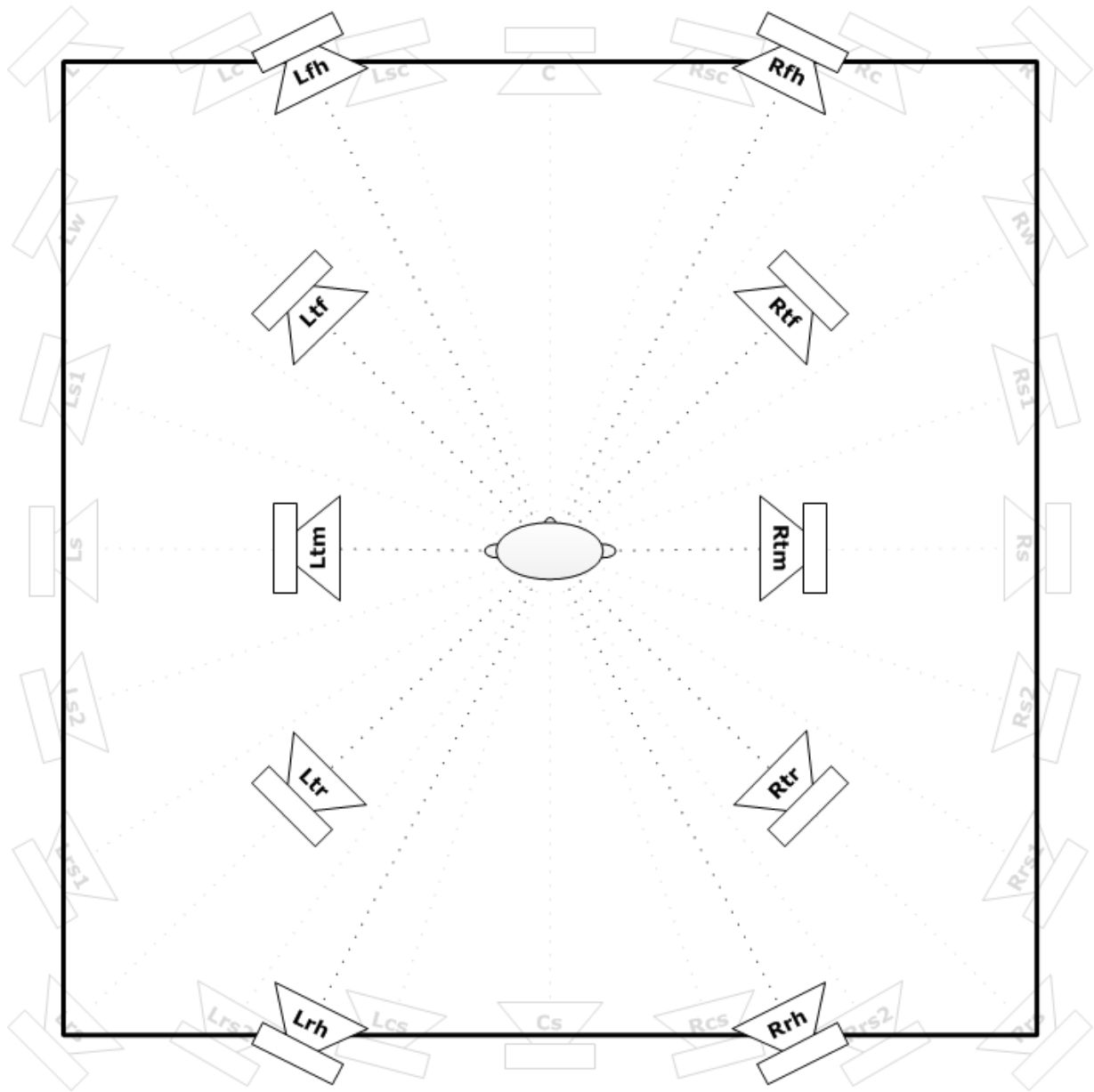


Figure 3-5 Relative Positions of Overhead Speakers (Not to Scale)

3.2.3 Device

Select the I/O device to be used for each channel. The decoder defaults to the built-in output of your computer. Only properly set up devices recognized by the system are listed.

3.2.4 Output

Select the output number of the selected I/O device to be used by the channel. Two channel IDs cannot be assigned to the same output channel.

3.2.5 Size

The **Size** setting selects whether to use bass management for the channel.

Large indicates that the speaker is considered full range and does not require bass management.

Small indicates that the speaker is a near-field or satellite design and is not designed to reproduce extremely low frequencies (typically, frequencies below 80 Hz). When a channel is set to **Small**, the system filters out low-frequency information from that channel and redirects it to subwoofer output. If no subwoofer is present in a listening mode, the selection of **Small** (or no bass management) is not allowed, and the **Size** field is grayed out. The subwoofer is required for greater than two-speaker layouts.

The **SW Crossover** selection in the lower part of the window determines the crossover frequency for bass management.

3.2.6 Trim

Use this control to trim output channels during sound system calibration. The range is quarter-decibel steps from 0 to –30 dB.

3.2.7 Delay

This setting controls the amount of delay (in milliseconds) that is added to the output channel for use in time aligning your sound system. (See [Calculating Speaker Delay Values](#) for more information.)

3.2.8 Subwoofer Crossover

The appropriate setting for the subwoofer crossover frequency depends on the capability of your front and surround speakers. When any channels are set to **Small** on the **Speaker Settings** page, the decoder will then divert signals below the subwoofer crossover value to the subwoofer output. Most smaller speakers require an 80 Hz subwoofer crossover, but you should use the setting most appropriate for your speakers. In a system with all speakers set to **Large**, the crossover value is not applied.

3.2.9 Dolby Pro Logic Delay

This option is disabled.

3.2.10 Testing a Listening Mode

Dolby Media Decoder makes it easy to test your listening modes (for example, to perform required speaker calibration).

For more information on speaker calibration, see [Chapter 10](#).

To test a mode:

1. Click the **Noise** button in the **Speaker Settings** page.

The system automatically cycles the noise through all of the channels present within the current listening mode.

2. If a channel needs level adjustment, click its **Trim** dial.
The system automatically locks the noise to the channel being trimmed.
3. Adjust other channels as required.
4. When you are satisfied with all of the channels, click the **Noise** button to turn noise off.

3.3 Changing Monitor Control Settings

The options on the **Monitor Control** page of the **Preferences** window determine the behavior of the controls in the monitor window. For example, the behavior of the **Custom** controls in the **COMPRESSION** section of the monitor window is controlled here, along with several other aspects of decoder behavior.



Figure 3-6 Monitor Control Page in the Preferences Window

There are three sections:

- **Dolby Pro Logic II** (options disabled)
- **Decode Controls** (options disabled)
- **Custom Compression**

The default settings for **Dolby Pro Logic II** and **Decode Control** are grayed out and cannot be changed.

The options on this page are the same as the options available in most consumer receivers. They allow you to emulate a broad range of consumer experiences in the studio.

For more information on the technology, see [Dolby Professional Technologies](#).

To change **Monitor Control** settings:

1. Choose **Dolby Media Decoder > Preferences**.
2. Click the **Monitor Control** tab.

The **Monitor Control** page displays.

3. Configure custom compression scale factors, as needed.

Settings are remembered when you move to another page, close the **Preferences** window, or close the application.

3.3.1 Dolby Pro Logic II

The options for Dolby Pro Logic® II are disabled.

3.3.2 Decode Controls

The options for **Decode Control** are disabled.

3.3.3 Custom Compression

The **Custom Compression** settings (**Boost** and **Cut**) determine the levels of boost and cut that are applied when you click the **Custom** button in the **COMPRESSION** section of the monitor window. This feature emulates a consumer receiver that allows customization of Line mode compression. Boost and cut can both be adjusted in tenths from 0.0 to 1.0. The default value for each is 1.0. For more information on compression algorithms, see [Compression Profile Buttons](#).

3.4 Dolby Headphone Output

The Dolby Headphone feature is disabled and its controls are grayed out.

Using Dolby Media Decoder

This chapter covers:

- [Using the Transport Window](#)
 - [Loading Audio Files to Play Back](#)
 - [Setting the Display Type](#)
 - [Setting the Display Mode](#)
 - [Setting Up Video for Synchronous Playback via a QuickTime Movie](#)
- [Setting Up Video for Synchronous Playback via a QuickTime Movie](#)
- [Using the Transport Controls](#)
- [Determining the Device Controlled](#)
- [Monitoring Playback](#)

4.1 Using the Transport Window

The transport window displays information about the loaded material and provides controls for playback.



Figure 4-1 Transport Window

The transport window includes the following displays and controls:

- **Information display:** Provides file name and other pertinent information, dependent on the type of file loaded. You can load audio or movies (.mov) files by dragging and dropping them into the display. You can also load audio by double-clicking in the display to display the **Open** window (where you can locate a file, highlight it, and then click the **Open** button).

The display includes:

- **File name:** Displays the name of the file (including its extension) on the first line.
- **Bitstream type:** Displays the Dolby® audio format (Dolby TrueHD, Dolby Digital Plus™, Dolby Digital, or MLP) on the second line. For bitstreams carrying Dolby

Atmos® audio and metadata, the bitstream type is followed by a **Dolby Atmos** indication.

The **Dolby Atmos** indication text is green for all bitstreams with Dolby Atmos objects and metadata. The indication is orange if the additional ADDBSI header information in Dolby Digital Plus is not present. The decoder supports playback for both signals.

- **(Downmix)** indication: Confirms Dolby TrueHD or MLP presentations that are downmixes. This indication is on the third line.
- **Dialogue**: Provides the dialogue normalization parameter on the right side of the information display.
- Data-rate indication: Displays the current data rate (in kbps) on the right side of the information display, below the **Dialogue** normalization parameter.
- Timecode counter: Displays timecode, based on the current display type and mode. The counter is optimized to be audio sample accurate when entering a time value.

When the display mode is set to **Source TC**, **Remote TC**, or **File Position**, the counter displays the current timecode. When the display mode is set to **File Start** or **File End**, the counter is fixed to the value for the respective mode. See [Section 4.2](#).

Timecode entries in the timecode counter are applied when window focus is changed (such as clicking outside the transport window or launching another window).

The timecode counter includes a playback media indicator (upper-right corner of the timecode counter): **A** for audio only, **V** for video only, or **AV** for audio/video.

- Timecode slider: Lets you change the position of playback material by dragging with a mouse. The slider is optimized to be audio sample accurate. When dragging, the following occurs:
 - The timecode counter shows the current position.
 - Video updates to the current position.
 - The meter display clears, and there is no sound.
- Transport controls: Provide controls for playback of audio, QuickTime® movies, and connected external devices. Controls include: first frame, rewind, stop, play, fast forward, and last frame. See [Section 4.3](#).
- Display type: Lets you set how the file position is displayed for both audio and QuickTime files. See [Section 4.1.2](#).
- Display mode: Lets you set which type of information is displayed in the counter. See [Section 4.1.3](#).
- **Remote** and **Sync** buttons: Let you set which devices the transport window will control. See [Section 4.3.1](#).

4.1.1 Loading Audio Files to Play Back

You can load and play back an encoded (multichannel interleaved) file with Dolby Media Decoder.

When loading Dolby TrueHD bitstreams, visual feedback of loading activity displays in the **Building Index for Audio File** window. You can press the **Cancel** button in this window to stop loading.

When a file is loaded, the file name and other pertinent information are displayed in the information display in the transport window.



Figure 4-2 Information Display in the Transport Window

Loading an Encoded File

You can load and play back an encoded file with Dolby Media Decoder.

To load an encoded file, do one of the following:

- Drag and drop an audio file onto the information display in the transport window.
- Double-click in the information display in the transport window. When the **Open** window displays, browse to the location of the file, highlight it, and click **Open**.
- Choose **File > Open File**. When the **Open** window displays, browse to the location of the file, highlight it, and click **Open**.
- Double-click an encoded file in the finder, if that extension has Dolby Media Decoder set as the default application.

4.1.2 Setting the Display Type

The value in the display type field determines how the file position is displayed for both audio and QuickTime files. The default display type is timecode indicated by a specific frame rate.



Figure 4-3 File Position Displayed in Timecode (23.976 fps)

The supported display types are:

- Timecode: 23.976, 24, 25, 29.97, and 29.97 df, and 30 fps.
- Real time.
- Frames.
- Seconds.
- Sample.

When an audio file is opened, the decoder automatically sets the display type to the frame rate of the timecode embedded within the file and displays the starting timecode. If no timecode is embedded in the file, the decoder switches to the real time value (in hour:min:sec:msec).

4.1.3 Setting the Display Mode

The value in the display mode field determines the type of information that will be displayed in the counter.

The supported display modes are:

- **Source TC:** Timecode embedded in the audio file. With **Source TC** as the display mode, the first timecode frame is the value embedded in the first frame of the file, which might or might not be **00:00:00:00**.
- **Remote TC:** Timecode obtained from a QuickTime movie if no audio is loaded, or if audio is loaded and the **Sync** button is off.
- **File Position:** File position according to the chosen display type with a zero reference start value. With **File Position** as the display mode, the first timecode frame is always **00:00:00:00**.
- **File Start:** The timecode embedded in the first frame of the file.
- **File End:** The timecode embedded in the last frame of the file.

4.2 Setting Up Video for Synchronous Playback via a QuickTime Movie

The decoder supports synchronizing video to audio via a QuickTime video file. Any video format supported by QuickTime may be used as long as it is in a .mov container. Refer to apple.com/quicktime for additional information on supported video formats.

When audio and video files begin or end at different times, the decoder starts playback at the earliest file and stops playback at the end of the last file to ensure that the full timeline is decoded.

The QuickTime movie screen includes a playback media indicator in the footer bar: **Audio**, **Video**, or **Audio/Video**.

To set up synchronization to a QuickTime movie:

1. From the menu bar, load the QuickTime movie, by doing one of the following:
 - Choose **File > Open Movie**, and then browse to the movie you want to open.
 - Drag and drop a QuickTime movie into the information display in the transport window.
 - Drag and drop a QuickTime movie into the QuickTime movie screen window.

2. The system opens the chosen file in the QuickTime movie screen window.



Figure 4-4 QuickTime Movie Screen



Tip: If your QuickTime movie is an anamorphic video, you can disable (unhighlight) the **Lock Aspect Ratio** button to unlock it. Unlocking the aspect ratio allows you to manually stretch the image for improved viewing.

3. Click the **TC Start** button in the lower-left corner of the window.
The system opens the **QuickTime Timecode Start** dialog.
4. In the timecode format drop-down menu, select the correct format.
5. Enter the correct **Timecode Start** (first frame of the video).
6. Click **OK** to accept the settings.



Tip: You must supply the correct frame rate and the correct value for the first frame of timecode for the QuickTime movie to allow synchronous playback with audio.

7. When setup is complete, click the **Sync** button in the transport window to activate synchronization.
Note: **Remote** and **Sync** buttons are enabled by default when you load audio and video.

4.3 Using the Transport Controls

You can use the transport controls in the transport window to control the playback of:

- Encoded Dolby audio files
- QuickTime–wrapped video files (.mov)

Table 4-1 Decoder Transport Controls

Icon	Function
	Home (first frame): Positions the decoder at the first frame of the open file. Keyboard shortcut: <Command>+<right arrow>.
	Rewind: Moves backward in time through the open file. Each time you click the button, movement accelerates. Keyboard shortcut: <left arrow>.
	Stop. Ends playback and returns the decoder to the first frame of the open file. Keyboard shortcut: <Enter>, which toggles Play/Stop.
	Play/pause. Keyboard shortcut: <Enter>, which toggles play/stop, or <space bar>, which toggles play/pause.
	Fast forward: Moves forward in time through the open file. Each time you click the button, movement accelerates. Keyboard shortcut: <right arrow>.
	End (last frame): Positions the decoder at the last frame of the open file. Keyboard shortcut: <Command>+<left arrow>.

4.3.1 Determining the Device Controlled

The state of the **Remote** and **Sync** buttons determines which devices the decoder will control.

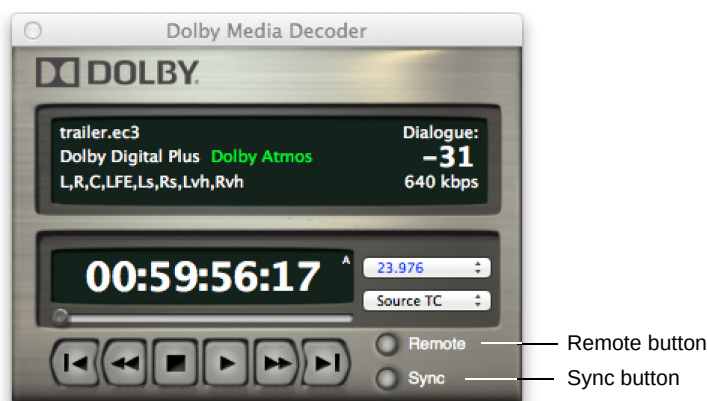


Figure 4-5 Remote and Sync Buttons

There are three possibilities:

- When neither button is lit, the transport controls the playback of loaded audio files only.
- When only the **Remote** button is lit, the transport controls the playback of QuickTime video files.
- When both the **Remote** and **Sync** buttons are lit, the transport controls both audio playback from Dolby Media Decoder and one video source simultaneously.

Click (highlight) a button to enable it. Click again to unhighlight it and turn it off. The **Sync** button is only functional when the **Remote** button is enabled. If you click the **Sync** button, the system will automatically enable the **Remote** button as well. The **Sync** button can then be switched off if desired to allow the transport to control the video only.



Note: **Remote** and **Sync** are enabled automatically when both an audio and a .mov file are loaded.



Note: Synchronous playback requires correctly specified timecode and frame rates for both audio and video. For details on the audio frame rates, see [Setting the Display Type](#). For details on the video frame rates, see [Setting Up Video for Synchronous Playback via a QuickTime Movie](#).

When audio and video files begin or end at different times, the decoder starts playback at the earliest file and stops playback at the end of the last file to ensure that the full timeline is decoded.

4.4 Monitoring Playback

Use the meters/gain window and monitor window to monitor playback levels.

4.4.1 Meters/Gain Window

The meters/gain window displays playback levels as well as the current channel configuration. Additionally, the window includes controls for changing output.

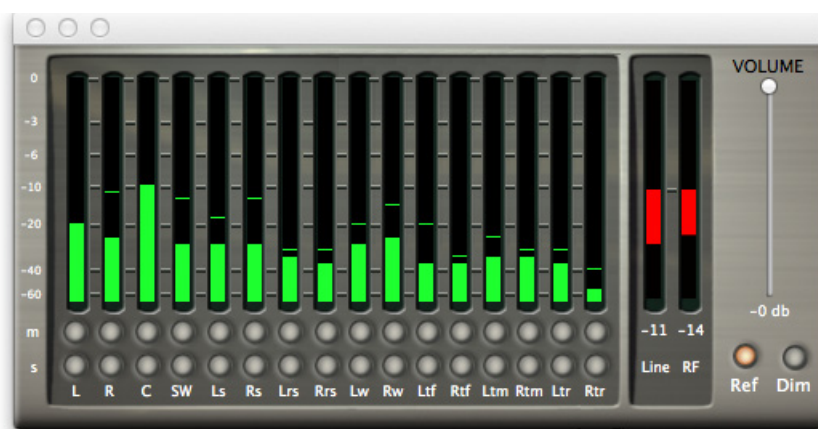


Figure 4-6 Meters/Gain Window with Updated Metering

The meters/gain window provides:

- **16-channel metering:** Supports speaker configurations of up to 16 speakers. Metering labels are at 0, -3, -6, -10, -20, -40, and -60 dB levels. Signal is displayed as follows: in green when below -6 dB; in yellow from -6 to -0.000005 dB; in red when at or above 0 dB.
- **Line and RF meters:** Display the cut or boost that would be applied if downmixing were to be enabled or if a late-night listening mode were selected (Line or RF mode). The scale for the RF boost and cut meter bar is compressed, because RF mode compression can vary over a wider range of values. When playing Dolby TrueHD files, Line mode and custom mode compression are available, but RF mode compression is not.
- **m and s buttons:** Let you mute or solo each channel, respectively.
- **Volume slider:** Adjusts the attenuation. The slider includes a digital display showing the attenuation value (in dBFS).
- **Ref button:** Stays lit as long as the master volume is 0 dB. If the slider is moved, the button becomes disabled. If you click the **Ref** button, the master volume is reset to 0 dB.
- **Dim:** Applies 20 dB of attenuation, when enabled.

4.4.2 Monitor Window

The monitor window provides controls for playing back downmixes, applying compression, and selecting which user-created listening mode (speaker settings) to use.

In Dolby Media Decoder v2.0, upmixing controls are grayed out and cannot be changed.

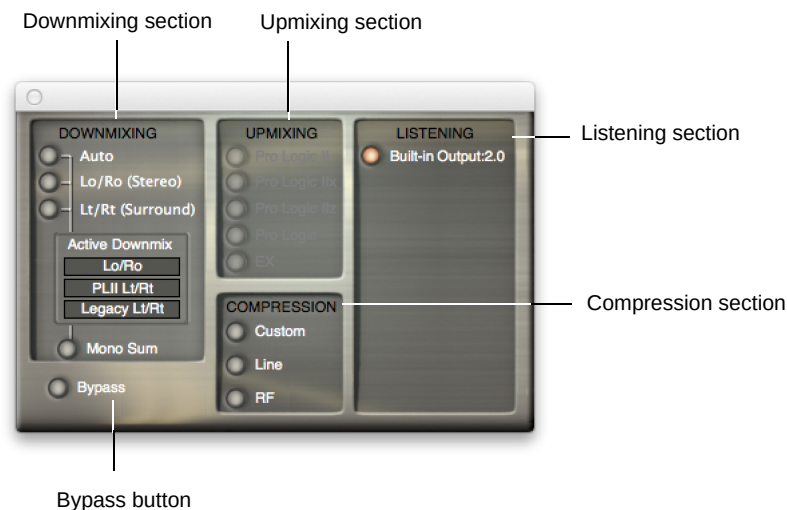


Figure 4-7 The Monitor Window

Bypassing Decoder Processing

You can bypass all processing for downmixing and dynamic range (except for dialnorm scaling) by clicking the **Bypass** button in the monitor window.

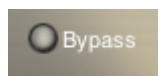


Figure 4-8 Bypass Button

When the decoder is in **Bypass** mode, none of the selected downmixing or compression will be heard. Dialogue normalization will still be applied. Settings on the **Speaker Settings** page are still active.

Auditioning Downmixes

When multichannel Dolby Digital and Dolby Digital Plus encoded files are loaded, you can listen to downmixes of the overall mix. The available choices are determined by the format.

The following modes are available with Dolby Digital and Dolby Digital Plus™:

- Auto
- Lo/Ro (Stereo)
- Lt/Rt (Surround)
- Mono Sum

Mono can be enabled for a two-channel downmix only when two-channel downmixing is active. We recommend that an Lo/Ro downmix is used before enabling Mono Sum mode. Use of an Lt/Rt downmix will result in loss of surround content during a mono summation.

To listen to a downmix of an encoded file:

- Click any of the buttons in the **DOWNMIX** section of the monitor window.
- Select a Dolby TrueHD presentation with a downmix in it.

Active Downmix

The **Active Downmix** section displays the actual downmix used for the current file. The specific type of downmixing applied will be determined by a combination of user settings, bitstream type, and bitstream metadata. In Auto mode, a flag in the bitstream may indicate that an Lo/Ro or an Lt/Rt downmix is preferred by the content creator. Dolby Digital Plus bitstreams, which are encoded with proper downmix overload protection for a Dolby Pro Logic® II compatible downmix, will have Dolby Pro Logic II Lt/Rt downmixing enabled when the user selects Lt/Rt mode. Dolby Digital Plus bitstreams that have a metadata flag set for legacy Lt/Rt as the preferred downmix, and all Dolby Digital bitstreams will have the legacy Lt/Rt downmix enabled when the user selects Lt/Rt mode.

Downmixes with Dolby TrueHD

With Dolby TrueHD, downmixing is handled by the encoder only. This differs from Dolby Digital and Dolby Digital Plus, where downmixing is handled by the decoder.

The Dolby TrueHD bitstream carries the downmixed audio in separate presentations. Therefore, when decoding Dolby TrueHD, the **DOWNMIX** section is replaced by a

Presentation section. This allows you to listen to either the two-, six-, or eight-channel presentation within a Dolby TrueHD stream. The **Presentation** section defaults to the largest independent presentation within the selected Dolby TrueHD bitstream, as follows.

- Dolby Atmos presentation is selected upon loading a Dolby TrueHD with Dolby Atmos file if the listening configuration is greater than 7.1 speaker channels.
- Eight-channel presentation is selected upon loading a channel-based Dolby TrueHD file if the listening configuration is greater than 5.1 speaker channels, up to 7.1 speaker channels.
- Six-channel presentation is selected upon loading any Dolby TrueHD file (with or without Dolby Atmos) if the listening configuration is greater than two speaker channels, up to six (5.1) speaker channels.
- Two-channel presentation is selected upon loading any Dolby TrueHD file (with or without Dolby Atmos) if the listening configuration has two speaker channels.

Auditioning Upmixes

The **UPMIXING** section options are disabled.

Compression Profile Buttons

The **COMPRESSION** section includes compression profiles, which can be applied to the Dolby TrueHD, Dolby Digital Plus, or Dolby Digital bitstream during decoding playback. These compression presets aid playback for two-channel downmix overload protection and late-night listening modes.

Compression profiles include:

- Custom: Emulates a consumer receiver that allows customization of Line mode compression during decoding playback. Boost and cut can both be adjusted in tenths from 0.0 to 1.0. The default value is 1.0. For information on creating a custom compression profile on the **Monitor Control** page of the **Preferences** window, see [Section 3.3](#).
- Line: Applies light compression characteristics during decoding playback. See [Line Mode](#).
- RF: Applies heavy compression characteristics during decoding playback. See [RF Mode](#).

When playing Dolby TrueHD files, Line mode and custom mode compression are available.

Line Mode

In Line mode, the normalized dialogue level is reproduced from the decoder at a constant loudness level of -31 dBFS Leq(A), assuming the dialogue level parameter is set correctly.

Line mode compression is designed to provide gentle, musical compression appropriate for the dynamic range expected from consumer line-level or power-amplified outputs from two-channel set-top decoders, two-channel digital televisions, 5.1-channel digital televisions, and Dolby Digital audio/video surround decoders.

Consumer control of the dynamic range is limited when downmixing. Products with stereo or mono outputs do not usually allow consumer scaling of Line mode. This is because these devices are usually downmixing (for example, when receiving a 5.1-channel signal).

However, in these products, the consumer may have a choice between Line mode and RF mode.

RF Mode

In RF mode, high- and low-level compression scaling is not allowed. When RF mode is active, that compression profile is always fully applied.

RF mode was originally designed for products (such as set-top boxes) that generated a downmixed signal for connection to the RF/antenna input of a television set; however, it is also useful in situations where heavy dynamic range compression is required (for example, when small PC speakers are used for DVD playback, for late-night listening, or anytime the user is in an environment with limited useful dynamic range or high noise levels). In RF mode, the overall program level is raised 11 dB. This results in dialogue being reproduced at an average level of -20 dBFS Leq(A), while the peaks are limited to prevent signal overload in the D/A converter. By limiting headroom, severe overmodulation of television receivers is prevented. The 11 dB gain provides an average loudness level that compares well with existing analog television broadcasts and is therefore used to match the levels of decoded TV content to other programs on a broadcast service.

For some types of speech-heavy content, it may be necessary to further constrain signal peaks above the average dialogue level so that there is less than 20 dB headroom. The selection of a suitable RF mode profile should be used during encoding.

Selecting a Listening Mode

You can select your own speaker settings and save them as a listening mode to use for playback monitoring. For information on creating and configuring a listening mode, see [Section 3.2](#).

To select a listening mode:

- Click a **Listening** mode button in the **LISTENING** section of the monitor window.

4.4.3 Buffer Underflow Indicator

The buffer underflow indicator turns red when Dolby Media Decoder is unable to get the input data from the loaded files quickly enough, or is unable to decode it fast enough to avoid gaps in the audio output.



Figure 4-9 Buffer Underflow Indicator in the Transport Window

If this indicator illuminates, we recommend freeing up system memory and CPU resources by closing other open applications, widgets, and any background processes that require CPU time or memory. In some cases, a faster hard-disk drive, faster network connection (for remote servers or remote file directories), or more computer RAM may be needed to ensure smooth, gapless audio playback of complex multichannel content.

Working with Decoder Configuration Files

Saving decoder configuration files—snapshots of the current state of the pages in the **Preferences** window—can eliminate repetitive selection of multiple settings. This can be particularly helpful when multiple users have their own preferred speaker settings. Once saved, decoder configuration files can be loaded with one click. There is no limit on the number of decoder configuration files you can save.

5.1 Saving, Loading, and Deleting Decoder Configuration Files

To save a new decoder configuration file:

1. Choose **Dolby Media Decoder > Preferences**.
2. Click the **Configurations** tab.

The **Configurations** page appears.

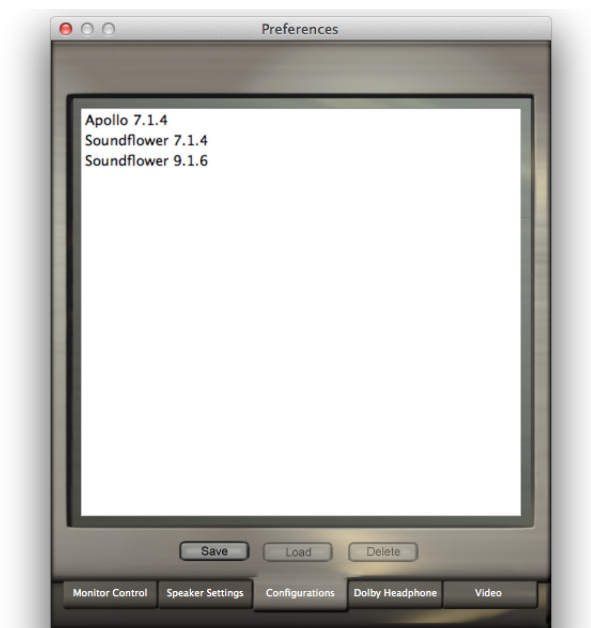


Figure 5-1 Example Configurations Page in the Preferences Window

The page lists all previously saved decoder configuration files and contains **Save**, **Load**, and **Delete** buttons.

3. Click **Save**.
The system opens a standard **Save** dialog.

4. Enter a name for the decoder configuration file. Then browse to a storage location, and click **Save**.



Note: We recommend saving the default state of the application as the first decoder configuration file.

We also recommend using clear, descriptive names for decoder configuration files to minimize search time.

To load a previously saved decoder configuration file, highlight its name in the list, then click **Load**.

To delete a previously saved decoder configuration file, highlight its name in the list, then click **Delete**.

Automatic Misconfiguration Detection

When you load a decoder configuration file, the decoder verifies the presence of the external I/O equipment saved in that configuration. If a component is missing, the system displays a warning.



Figure 5-2 Example Misconfiguration Warning

Missing devices appear in red on the **Speaker Settings** page.



Figure 5-3 Missing Device Indications on the Speaker Settings Page

Viewing Metadata

You can view the metadata for an open file in the metadata window.

6.1 The Metadata Window

The metadata window displays file information and metadata for the current open file. Each type of encoded file contains a different set of metadata parameters, so this window will look different depending on which file type is loaded.

Metadata in this window is read only. However, metadata for Dolby® audio formats without Dolby Atmos® can be edited in Dolby Media Tools. See the *Dolby Media Tools User's Manual*.



Figure 6-1 Metadata Window for Dolby Digital Plus Bitstream with Dolby Atmos

The metadata window headers display the type of encoded file. For example, **Dolby Digital Plus w/ Dolby Atmos** displays for Dolby Digital Plus™ with Dolby Atmos encoded files.

The **Bitstream Type** field in the metadata field notes whether the encoded file bitstream is Dolby Atmos or channel based.

For Dolby TrueHD with Dolby Atmos or Dolby Digital Plus with Dolby Atmos encoded files, the metadata section includes the number of **Encoded Elements** (the number of

elements set during encoding with the Dolby Media Encoder) and the **Dolby Atmos ADDBSI Flag**. The **Dolby Atmos ADDBSI Flag** value is **Yes** when a Dolby Digital Plus file contains the additional header data that verifies that the bitstream is a Dolby Atmos bitstream.

6.2 Viewing Metadata in the Metadata Window

You can view the metadata for an open file in the metadata window.

To open the metadata window (or close when open), do one of the following:

- From the menu bar, choose **Window > Metadata**.
- With the decoder menu in focus, press <M> on the keyboard.

Application Menus

This chapter describes the menus in the Dolby® Media Decoder interface:

- [Dolby Media Decoder Menu](#)
- [File Menu](#)
- [Edit Menu](#)
- [Transport Menu](#)
- [Monitoring Menu](#)
- [Window Menu](#)
- [Help Menu](#)

7.1 Dolby Media Decoder Menu

The **Dolby Media Decoder** menu contains commands for managing the application window.

Dolby Media Decoder	File	Edit
About Dolby Media Decoder		
Preferences...	⌘,	
Services		▶
Hide Dolby Media Decoder	⌘H	
Hide Others	⌘H	
Show All		
Quit Dolby Media Decoder	⌘Q	

Figure 7-1 Dolby Media Decoder Menu

These include:

- **About Dolby Media Decoder:** Displays version and authorship information.
- **Preferences:** Opens the **Preferences** window, with dedicated pages for **Monitor Control**, **Speaker Settings**, **Configurations**, **Dolby Headphone**, and **Video**.
- **Services:** Lists all of the available machine services. This is a Standard Mac® command.
- **Hide Dolby Media Decoder:** Hides the **Dolby Media Decoder** window without closing the application.
- **Hide Others:** Hides all open windows from other applications without closing them.
- **Show All:** Shows all applications that were hidden after using any of the menu **Hide** commands.
- **Quit Dolby Media Decoder:** Quits the application.

7.2 File Menu

The **File** menu contains commands for opening and closing files.

File	Edit	Transport	Moni
Open File...			⌘O
Open Source Files...			⌘S
Open Movie...			⌘M
Open Recent			▶
Close			⌘W
Print Metadata...			⌘P
Decode To Files...			⌘D

Figure 7-2 File Menu

These include:

- **Open File:** Opens audio encode files or interleaved multichannel audio files (.wav or .aif).
- **Open Source Files:** This feature is disabled.
- **Open Movie:** Opens a QuickTime® video file.
- **Open Recent:** Opens a recent file from the submenu.
- **Close:** Closes the open audio file.
- **Print Metadata:** Prints the metadata for the current file. By default, the default printer of the local machine is selected. You can also save the file in PDF format.
- **Decode to Files:** Creates 24-bit multi-mono .wav files decoded from the encoded file that is loaded. When a Dolby TrueHD file is loaded, the files that will be created are based on the presentation that is selected on the interface (eight, six, or two channel). The decoded file names are automatically generated from the name of the encoded file that is loaded. Dolby Media Decoder supports decoding presentations up to 7.1 from a Dolby Atmos® bitstream.

4 GB and larger files are decoded to the RF64 standard.

7.3 Edit Menu

The **Edit** menu includes basic commands for managing text in text fields

Edit	Transport	Monitoring	Wi
Cut			⌘X
Copy			⌘C
Paste			⌘V
Start Dictation			fn fn
Special Characters...			^⌘Space

Figure 7-3 Edit Menu

These include:

- **Cut:** Removes highlighted text in a text field.
- **Copy:** Copies highlighted text in a text field.

- **Paste:** Adds the previously cut or copied text into the selected text field (or another application).
- **Special Characters:** Provides access to symbols and other special characters for use in text fields.

7.4 Transport Menu

The **Transport** menu includes commands for transport control. Commands are listed with their keyboard shortcuts, where available.

Transport	Monitorir
Play/Stop	↵
Play/Pause	Space
Home	⌘←
End	⌘→
Rewind	←
Forward	→
Dim	
Ref	⌘Z

Figure 7-4 Transport Menu

These include:

- **Play/Stop** <Enter>: Plays the current file if it is stopped, or stops the file if it is playing.
- **Play/Pause** <space bar>: Plays the current file if it is paused, or pauses the file if it is playing.
- **Home** <Command>+<left arrow>: Returns to the beginning of the current file.
- **End** <Command>+<right arrow>: Advances to the end of the current file.
- **Rewind** <left arrow>: Moves backward through the current file.
- **Forward** <right arrow>: Moves forward through the current file.
- **Dim**: Reduces the volume by 20 dB.
- **Ref** <Command>+<Z>: Returns volume to 0 dB.

7.5 Monitoring Menu

The **Monitoring** menu lets you choose from options that correspond to the **Downmixing**, **Bypass**, **Upmixing**, **Compression**, and **Listening** options listed in the monitor window. For more information, see [Section 4.4.2](#).

With Dolby Media Decoder v2.0, the Dolby Pro Logic® II, Pro Logic, and Dolby Digital Surround EX™ options are grayed out and cannot be changed.

Monitoring	Window	Help
Auto		Q
Lo/Ro (Stereo)		W
Lt/Rt (Surround)		^W
Mono		R
Bypass		⌘B
Pro Logic II		A
Pro Logic		S
EX		D
Custom Compression		Z
Line Compression		X
RF Compression		C
✓ Built-in Output:2.0		⌘1

Figure 7-5 Monitoring Menu

7.6 Window Menu

The **Window** menu contains commands for managing the application windows. Key shortcuts are provided, where applicable.

Window	Help
✓ Transport	
✓ Monitor	
✓ Meters/Gain	
Metadata	M
QuickTime	V
Minimize	
Zoom	
Bring All to Front	
Latch	
Latch All	
Unlatch	
Unlatch All	
Dolby Media Decoder	

Figure 7-6 Window Menu

These include:

- **Transport:** Shows (or hides) the transport window.
- **Monitor:** Shows (or hides) the monitor window.
- **Meters/Gain:** Shows (or hides) the meters/gain window.
- **Metadata:** Shows (or hides) the metadata window. The keyboard shortcut is <M>.
- **QuickTime:** Opens (or hides) the QuickTime movie screen. The keyboard shortcut is <V>.
- **Minimize:** This menu command is disabled.
- **Zoom:** This menu command is disabled.
- **Bring All to Front:** Moves all the open application windows to the front.

- **Latch:** Attaches two open windows along a vertical edge.
- **Latch All:** Attaches the three main windows in their default order with the transport window on the left, the meters/gain window in the center, and the monitor window on the right.
- **Unlatch:** Allows each of the selected windows to be moved around the screen independently.
- **Unlatch All:** Allows all of the windows to be moved around the screen independently.

7.7 Help Menu

The **Help** menu provides access to a .pdf file of the *Dolby Media Decoder User's Manual*.

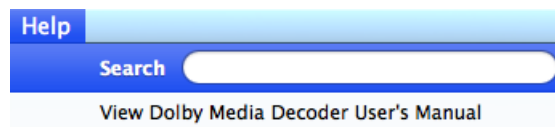


Figure 7-7 Help Menu

Dolby Media Decoder Frequently Asked Questions

This chapter answers the following questions:

- [How Can I Get My Pro Tools I/O to Appear in the Output Device Selection Drop-Down Menu on the Speaker Settings Page?](#)
- [How Do I Get Audio from All of the Channels of My Pro Tools I/O?](#)
- [Why Does Dolby Media Decoder Not Recognize My I/O Device?](#)
- [Why Am I Not Hearing Audio?](#)
- [Why Does the Audio Sound Incorrect?](#)

8.1 How Can I Get My Pro Tools I/O to Appear in the Output Device Selection Drop-Down Menu on the Speaker Settings Page?

Make sure that neither Avid® Pro Tools® nor any other program is currently running and accessing the I/O. The decoder cannot access an I/O device being used by any other program. For more information, see [Setting Up Internal or External Audio Hardware](#).

8.2 How Do I Get Audio from All of the Channels of My Pro Tools I/O?

Make sure the channel mapping on your Pro Tools or other I/O device is correct. The setup programs for Avid and other interfaces can alter the analog, AES, and S/PDIF output assignments.

8.3 Why Does Dolby Media Decoder Not Recognize My I/O Device?

Dolby® Media Decoder requires Core Audio–compliant I/O. Most I/O devices require the installation of Core Audio or ASIO drivers on your computer. See your I/O manufacturer's specification for more information. Many I/O manufacturers offer driver downloads from their websites or include them on the installation discs shipped with the products. For more information, see [Setting Up Internal or External Audio Hardware](#).

8.4 Why Am I Not Hearing Audio?

The most common causes of nonexistent audio are lack of power and faulty physical connections. Verify the following:

- Your computer is connected to a supported I/O device.
- You have downloaded and installed all required drivers.
- You have configured speakers on the **Speaker Settings** page.
- All devices requiring independent power are plugged in and receiving power.
- All devices with power switches are switched on.
- All devices in the chain are properly connected.
- The cables work.



Note: The best practice is to start at one end of the device chain and verify each connection in order.

8.5 Why Does the Audio Sound Incorrect?

The most common cause of warbling or other audio anomalies is a mismatch between any of several recording and playback parameters. Among the common mismatches are:

- Playing a source back at an incorrect frame rate
- Playing a source back at an incorrect sample rate
- A clocking mismatch between the digital audio devices

Any of these can also cause pops and clicks, which are often particularly noticeable at editing points.

Dropouts are commonly the result of compromised input files. For channel-based encodes created from .wav or .aif source files, check the original .wav or .aif files at the time points where dropouts were observed in the encoded file. Use a digital audio workstation or other PCM playback software to check the files. For Dolby Atmos–based encodes, review the source .atmos master file. Use the Dolby Rendering and Mastering Unit for Home Theater with the **Read-only Playback** option enabled to play back the master.

The following table describes some of the most common single-detail faults.

Table 8-1 Single-Detail Problems and Diagnoses

Problem	Diagnosis
The program sounds fine, but the dialogue is missing.	This is usually the result of replaying a 5.1-channel program over just two speakers without downmixing.
Sound is coming from the wrong speaker (dialogue from a surround speaker, for example).	The output channels are likely improperly mapped in the decoder. Another possibility is that speaker wires are improperly connected.
The sound is too compressed.	Compression is on when it is not needed.

Reference Information

This chapter provides reference information on Dolby® professional technologies.

For reference information on encoding options for Dolby TrueHD, Dolby Digital, Dolby Digital Plus™, and MLP Lossless™ formats, see the *Dolby Media Encoder User's Manual*.

9.1 Dolby Professional Technologies

The Dolby Media Decoder supports all of the following Dolby audio technologies:

- [Dolby Atmos](#)
- [Dolby TrueHD](#)
- [Dolby Digital Plus](#)
- [Dolby Digital](#)
- [MLP Lossless](#)

9.2 Dolby Atmos

With Dolby Atmos®, sound comes alive from all directions, including overhead. The Dolby Atmos format combines a channel-based audio bed with object-oriented sound to place and move specific effects around the listener. Using dynamic audio objects, sound designers and artists are free to mix audio in a three-dimensional space, steering effects throughout the room. When the Dolby Atmos soundtrack is played back in the cinema, the object-based audio mix is adapted to the room. (That is, the audio is rendered to the exact speaker layout of the theatre.)

A major benefit of Dolby Atmos is the ability to scale a soundtrack to different room sizes and speaker configurations. Dolby Atmos provides an optimal presentation of the soundtrack for each listening environment.

Audio/video receiver and home-theater-in-a-box products were the first consumer products to support Dolby Atmos. These products decode, scale, and calibrate Dolby Atmos content to accommodate a variety of home theater configurations. The initial Dolby Atmos products can support as few as 7 speakers and, for certain products, as many as 34 speakers.

Dolby Atmos for home theater has the following features to produce an incredible, enveloping, three-dimensional audio experience:

- Object-based Dolby Atmos content
- Dolby Atmos decoders
- Dolby advanced audio postprocessing
- Dolby Atmos enabled speaker technology

9.3 Dolby TrueHD

Dolby TrueHD is Dolby's lossless technology for high-definition disc-based media. Dolby TrueHD delivers sound that is bit-for-bit identical to the studio master, unlocking the true high-definition entertainment experience on current and future surround formats on Blu-ray Disc. When coupled with high-definition video, Dolby TrueHD offers an unprecedented home theater experience that lets you enjoy sound as stunning as the high-definition picture.

9.3.1 Features

The most notable features of Dolby TrueHD are:

- Lossless coding technology
- Supports carriage of Dolby Atmos
- Supports data rates up to 18 Mbps
- Supports up to eight full-range channels of 24-bit/96 kHz audio and up to six channels of 24-bit/192 kHz audio
- Supported by High-Definition Media Interface™ (HDMI™)
- Supports extensive metadata, including dialogue normalization and dynamic range control
- Supports multiple discrete presentations that can be custom mixes for Dolby Atmos, 7.1, 5.1, or stereo.

9.3.2 Benefits

The most notable benefits of Dolby TrueHD are:

- Studio-master-quality sound that unlocks the true high-definition entertainment experience on next-generation discs.
- Compatible with the A/V receivers and home-theater-in-a-box products of today and tomorrow.
- Dialogue normalization that can maintain the same volume level when you change to other Dolby TrueHD, Dolby Digital Plus, and Dolby Digital programming.
- Dynamic range control (Night mode) enables you to customize audio playback to reduce peak volume levels while experiencing all the details in the soundtrack.
- Selected as a format for Blu-ray Disc™.

9.3.3 Compatibility with Current A/V Receivers

Whether you own an older A/V system equipped with Dolby Digital, or a newer A/V receiver equipped with an HDMI connection or external line-level multichannel inputs, you are ensured a compatible, high-quality playback path.

9.4 Dolby Digital Plus

Dolby Digital Plus is the next-generation audio technology for all high-definition programming and media. It combines the efficiency to meet future broadcast demands with the power and flexibility to realize the full audio potential of the high-definition experience. Built on Dolby Digital, the multichannel audio standard for DVD and HD broadcasts worldwide, Dolby Digital Plus was designed for the delivery formats of the

future, but remains fully compatible with all current A/V receivers. With Dolby Digital Plus, you get even higher quality audio, more channels, and greater flexibility.

9.4.1 Features

The most notable features of Dolby Digital Plus are:

- Multichannel sound with discrete channel output.
- Supports carriage of Dolby Atmos
- Channel and program extensions can carry multichannel audio programs of up to 7.1 channels and support multiple programs in a single encoded bitstream.
- Outputs a Dolby Digital bitstream for playback on existing Dolby Digital systems.
- Supports data rates as high as 6 Mbps.
- Bit-rate performance of up to 1.6 Mbps on Blu-ray Disc.
- Accurately reproduces what the director and producer intended.
- Interactive mixing and streaming capability in advanced systems.
- Supported by High-Definition Media Interface (HDMI).

9.4.2 Benefits

The most notable benefits of Dolby Digital Plus are:

- Can deliver 7.1 channels or more enhanced-quality audio at up to 6 Mbps
- Allows multiple languages to be carried in a single bitstream
- Offers audio professionals new creative power and freedom
- Compatible with the millions of home entertainment systems equipped with Dolby Digital
- No latency or loss of quality in the conversion process; maintains high quality at more efficient broadcast bit rates (<192 kbps for 5.1-channel audio)
- Selected by the Advanced Television Systems Committee (ATSC) as the standard for future broadcast applications; named as an option by the Digital Video Broadcasting (DVB®) Project for satellite and cable TV
- Selected as a format for Blu-ray Disc

Next-generation high-definition disc players are compatible with current audio/video (A/V) receivers. Whether you own an older A/V system equipped with Dolby Digital, or a newer A/V receiver equipped with an HDMI connection or external line-level multichannel inputs, you are ensured a compatible, high-quality playback path.

Dolby Digital Plus can support more than eight audio channels, even though Blu-ray Disc standards currently limit their maximum number of audio channels to eight.

9.5 Dolby Digital

In thousands of cinemas and millions of homes worldwide, Dolby Digital is the standard for surround sound technology in general and 5.1-channel surround sound in particular.

Dolby Digital is a highly sophisticated and versatile audio encoding/decoding technology. Dolby Digital technology can transmit mono, stereo (two channel), or up to 5.1-channel surround sound (discrete multichannel audio).

The superior coding efficiency of Dolby Digital—and its ability to deliver high-quality discrete multichannel audio without compromising video quality—has made it the designated audio standard for DVD worldwide.

Dolby Digital is also the preferred multichannel audio standard for broadcast direct to home satellite and digital cable systems. Nearly 60 million digital cable and satellite set-top boxes are currently in use worldwide, supported by over 30 million audio/video receivers equipped with 5.1-channel Dolby Digital decoding. Virtually every DVD-Video player sold today offers a two-channel downmix of the Dolby Digital 5.1-channel movie soundtrack, as well as a digital output for connection to a 5.1-channel audio/video receiver.

In the film industry, the Dolby Digital soundtrack is optically encoded right on the filmstrip, in the space between the sprocket holes. Placing the soundtrack directly on the film allows it to coexist with the analog track, a cost-effective and simple solution for film distributors and theatre owners. The sprocket hole area has also proven highly resistant to wear and tear, so that a soundtrack encoded in Dolby Digital will remain free of pops and hiss for the useful life of the print.



Note: The sound information contained in each of the six available channels is distinct and independent. These six channels are described as a 5.1-channel system, because there are five full-bandwidth channels (Left, Right, Center, Left Surround, and Right Surround) with a 3 Hz to 20 kHz frequency range, plus a Low-Frequency Effects (LFE) subwoofer channel devoted to frequencies from 3 to 120 Hz.

9.6 MLP Lossless

MLP Lossless, licensed by Dolby Laboratories, enables producers to encode up to six channels of 24-bit/96 kHz audio, or two channels of 24-bit/192 kHz audio, resulting in playback that is bit-for-bit identical to the studio master. Nothing is lost during the encoding and decoding process.

Calibrating the Speaker System

Prior to decoding audio, you must calibrate the monitoring system to establish a balance between all channels and to ensure that all speakers play back at the correct level relative to the listening reference position.

10.1 Speaker Calibration Functionality

The built-in speaker calibration provides specific functionality as you enable noise, calibrate noise, and modify trims and delay controls during calibration.

Consider the following before you enable noise for speaker calibration:

- Initial speaker calibration:
 - It is your responsibility to save the listening layout before using it for speaker calibration.
 - If the **Speaker Settings** page contains unsaved changes, you will not be able to start calibration. An error window displays with following text: **This listening layout contains unsaved changes. Please save your changes and restart the noise generation.**
 - Clicking the **Noise** button starts the current listening layout displayed at the top of the **Speaker Settings** page in the **Preferences** window.

If the selected listening mode is broken or not set, an error message displays and the noise generation stops.

 - Calibration will always initially start in an automatic mode. In this mode, noise continuously switches through each speaker, in order of their presence in the meter display.
- Decoder functionality during noise calibration:
 - Transport controls (play, stop, fast forward, and rewind) are ignored.
 - Opening or closing a file is not allowed.
 - The small **C** timeline state indicator displays (instead of **A**, **V**, or **AV**).
 - Mute and solo buttons are disabled.
 - **Downmixing**, **Upmixing**, **Bypass**, **Compression**, and presentation controls remain disabled.
 - **Listening** mode in the **Preferences** window and **Listening** layout in the monitor window are linked.
 - Switching **Listening** mode in the **Preferences** window or **Listening** layout in the monitor window restarts the calibration procedure from the beginning of that listening layout. If the listening layout is invalid or the audio interface reports an error, an error message displays and the noise generation stops.
 - Switching the listening mode layout does not save changes made to **Trim** or **Delay**. You must manually save the changes by pressing the **Save** button.
 - Pressing **Save** stops calibration and saves the listening layout.
 - Pressing **Delete** stops calibration and removes the layout.

- Bass management will not affect noise generation, except for noise on the subwoofer output, which is passed via a lowpass filter on the selected cutoff frequency.
- Controls and options in all of the pages of the **Preferences** window are disabled. These include **Monitor Control**, **Configurations**, **Dolby Headphone**, and **Video**.
- Trim and delay modifications during noise calibration:
 - The modified value is applied live.
 - Switch calibration changes from its automatic mode to its manual mode. In this mode, noise is constantly generated on the speaker where the modification occurs.
 - To switch to another speaker, modify either the **Trim** or **Delay** value for the respective speaker.
 - To return calibration to its automatic mode, stop calibration and start it again.
 - Modified **Trim** or **Delay** values are not saved during calibration. You must manually save the changes by pressing the **Save** button.

10.2 Calibrating Speaker Playback Levels

In order to calibrate speaker playback level, you will need to choose how you will adjust system playback levels. You will also need to measure sound pressure level (SPL), plan subwoofer calibration, and calculate speaker and Dolby® Pro Logic® delays.

There are three options to adjust monitor system playback levels:

- Amplifier gain trim controls
- Mixer group outputs (one for each of the Left, Center, Right, Left Surround, Right Surround, and Low-Frequency Effects channels)
- Decoder output-level trim controls

The best option is to use your amplifier gain controls to set proper playback levels. This lets you maintain optimum signal-to-noise performance from the decoder and console.

When adjusting playback levels via console group outputs, pink noise readings depend on the type of console meter used to set reference level (and, strictly speaking, on the bandwidth of the pink noise signal). In film practice, pink noise level is read with a true VU meter, or a meter display with a VU characteristic. If reference level is specified as 0 VU, where 0 VU corresponds to –20 dBFS for a digital recording medium, then pink noise should be set to 0 VU on the console meter, and the SPL set accordingly.



Note: Many modern consoles have peak-reading meters or meter displays. Pink noise that reads at reference level using a true VU meter will read from 8 to 12 dB higher on a peak-reading meter or meter display. If your console has switchable meter characteristics, be sure to choose VU mode before setting pink noise levels.

To properly calibrate speaker levels, use an SPL meter. Suitable and relatively inexpensive meters are widely available. Relative level between channels is more important than absolute level, so the accuracy of this meter is sufficient for channel balancing.

What is most important is that all of the main channels are set to the same SPL—the absolute level is secondary. However, as a guide, the following are some examples of conventional level settings.

For film work, test noise at reference level should produce an SPL of 85 dBC.

For television work, test noise at reference level is typically set to produce an SPL ranging from 79 to 82 dBC for each of the five main channels. The lower reference level for television reflects the lower average listening levels preferred by the consumer (typically 70 to 75 dBC).

For music mixing, as for television, each speaker channel should be set to the same SPL. There is no standard reference level for music mixing. Some engineers prefer to mix louder than others do, but if the levels between channels are correct, the overall level is not as crucial.

When mixing for television or music in very small mixing rooms (for example, remote recording trucks), the surround channels are sometimes set 1 to 2 dB lower than the front channels. This takes into account the short distance to the surround speakers. Experience has shown that this setting makes the sound heard by the mixer more accurately reflect the sound heard in the home environment.

10.2.1 Measuring SPL

Before generating test noise, press the **Ref** button on your SPL meter to check that the master volume level is set to 0 dB. Master volume should remain at 0 dB while you set channel trim levels. The **Ref** button remains lit as long as the master volume level is set to 0 dB. If the light turns off, press the **Ref** button again to return the level to 0 dB. If the overall level needs adjustment when you begin measuring SPL, adjust the master trim level before adjusting the levels for individual channels.



Note: Before you turn on a test noise, be sure that your playback system is set to a moderate listening level. Adjust your amplifiers, self-powered speakers, or mixer. Do not adjust the master volume. Be aware that if the playback level is very high, you may risk damaging your speakers or possibly your hearing. The default test noise level is -20 dBFS to the digital outputs (+4 dBu to the analog monitor outputs).

To generate test noise to the analog outputs, click **Noise** on the **Speaker Settings** page in the **Preferences** window.

The system automatically cycles the noise through all of the channels present within the current listening mode. Test noise remains two seconds at each output before moving on to the next.

When you enter an individual channel trim adjustment menu, the test noise remains at that channel until you complete the adjustment and move to another channel, which allows you to adjust the SPL for each channel carefully, using the following procedure.

To accurately measure SPL:

1. Sit in the reference listening position. Set the SPL meter to C weighting and slow response.
2. Facing the front speakers, hold the SPL meter at chest level, with the microphone facing up at an angle of approximately 45 degrees to the Center speaker. Keep the meter at arm's length to prevent measuring audio that may reflect from your body.

3. Keep the SPL meter in this position. Make sure that the meter is aimed at the Center speaker as you take readings for the Left, Center, and Right speakers.
4. To take the SPL readings for the Left Surround or Right Surround speakers, keep the meter at the same angle and position as you did for the front speakers. Turn your body 90 degrees from the Center speaker toward the wall closest to the surround speaker you are measuring. This minimizes shadowing (obscuring the meter with your body).

To adjust an individual channel trim setting, dial a number using the up or down arrow.

10.2.2 Subwoofer Calibration

The ideal test noise for subwoofer calibration should be band-limited pink noise, lowpass filtered at 120 Hz. The decoder outputs band-limited pink noise (20 to 120 Hz) on the Low-Frequency Effects channel output. To properly calibrate the subwoofer, a real-time analyzer (RTA) is required. If an RTA is not available, you can approximate the settings with an SPL meter.

When using an RTA, proper calibration requires setting the Low-Frequency Effects channel signal to be sent to the subwoofer, within its typical bandwidth of 20 to 120 Hz, 10 dB higher (as measured by the RTA) than the main channels. [Figure 10-1](#) is an example of the RTA display of a properly calibrated subwoofer. The precision of this measurement varies with the quality of the meter used.

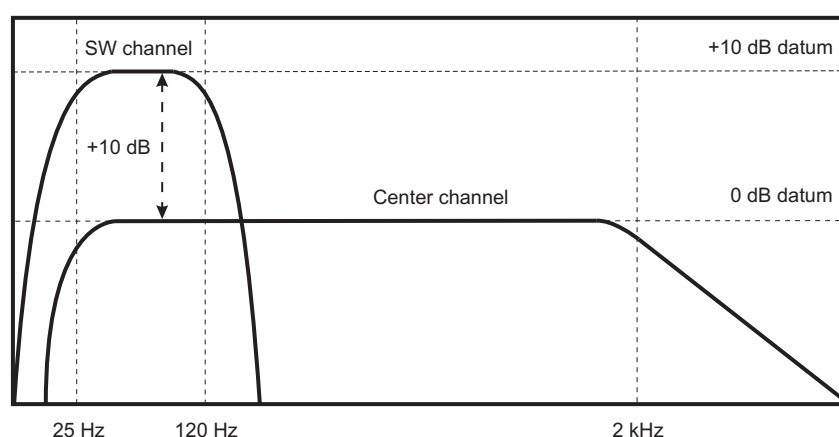


Figure 10-1 Real-Time Analyzer Display Simulation

If an RTA is not available, setting the Low-Frequency Effects channel 4 to 6 dB higher, as measured by an SPL meter, provides an approximate level. For example, set the subwoofer to 89 dBC when the Center channel measures 85 dBC.

For future reference, if you calibrate the subwoofer with an RTA, measure the calibrated level with an SPL meter and note the meter reading for the proper calibration. Use this measured value for quick future checks of the system calibration.

10.3 Calculating Speaker Delay Values

Adjusting speaker delay values is an essential step in configuring your system. The delay values for listed channels should be calculated for coincident arrival (so that signals from all speakers arrive at the reference listening position at the same time).

To calculate the delay settings, measure and record the distance from the reference listening position to each speaker. Use the longest distance from any channel to the reference listening position as the benchmark, and calculate delays in reference to it.



Note: This is a change from previous Dolby recommendations that used the equidistant Left and Right channels as the benchmark. This change is made to allow greater flexibility in positioning speakers.

Assign the furthest channel 0 delay, and add 1 ms delay to each channel for each foot closer they are to the reference position. For example, if the distance to the Right speaker is ten feet, the distance to the Center speaker is eight feet, and the distance to the Left speaker is six feet, assign a 2 ms delay to the Center channel, and a 4 ms delay to the Left channel.

All delays must be positive. The default is 0.